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31 March 1972
SUPERSEDING
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(See 6.6)

MILITARY SPECIFICATION

COMBINER, EXTENDED FREQUENCY RANGE CU-2007( )/SRR

1. SCOPE

1.1 This specification covers the requirements for an extended frequency range combiner CU-2007( )/SRR that is intended to permit existing 2 Megahertz (MHz) to 30 MHz shipboard antennas to be used with receivers having operating frequencies below 2 MHz.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-S-901	- Shock Test, H.I. (High-Impact), Shipboard Machinery, Equipment and Systems, Requirements for (NAVY)
MIL-E-16400	- Electronic Equipment, Naval Ship and Shore, General Specification
MIL-E-17555	- Electronic and Electrical Equipment, Accessories, and Repair Parts, Packaging and Packing of
MIL-F-19207	- Fuseholders, Extractor Post Type, Blown Fuse Indicating and Nonindicating, General Specification for
MIL-T-23103	- Thermal Performance Evaluation, Airborne Electronic Equipment, General Requirement for
MIL-I-45208	- Inspection System Requirements

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-108	- Definitions of, and Basic Requirements for Enclosure for Electric and Electronic Equipment
MIL-STD-109	- Quality Assurance Terms and Definitions
MIL-STD-129	- Marking for Shipment and Storage
MIL-STD-189	- Racks, Electrical Equipment, 19-Inch and Associated Panels

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MIL-STD-415	- Test Provisions for Electronic Systems and Associated Equipment, Design Criteria for
MIL-STD-461	- Electromagnetic Interference Characteristics, Requirements for Equipment
MIL-STD-470	- Maintainability Program Requirements (for Systems and Equipments)
MIL-STD-471	- Maintainability Demonstration
MIL-STD-781	- Reliability Tests Exponential Distribution
MIL-STD-785	- Reliability Program for Systems and Equipment Development and Production
MIL-STD-1345	- Data, Measurement, in Support of Maintenance, Calibration, and Repair of Electronic Equipment
MIL-STD-1364	- Preferred General Purpose Electronic Test Equipment
MIL-STD-1472	- Human Engineering Design Criteria for Military Systems, Equipment, and Facilities

HANDBOOKS

MILITARY

MIL-HDBK-217	- Reliability Stress and Failure Rate Data for Electronic Equipment
MIL-HDBK-472	- Maintainability Prediction

PUBLICATIONS

MILITARY

NAVSHIPS 0967-303-8610	- Technical Manual for Antenna Coupler Groups AN/SRA-38, 39, 40, 49, and 50
NAVSHIPS 0967-320-0010	- U.S. Navy Installation Criteria for CY-4516(A)/S Shipboard Electronic Equipment Cabinet

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General. The equipment to be supplied shall meet all the applicable detail requirements of MIL-E-16400, except as hereinafter specified.

3.1.1 First article sample. Prior to beginning production a sample shall be tested as specified in 4.3 (see 6.4). The sample shall be suitable for complete evaluation of mechanical and electrical form, design, and performance. It shall be of final mechanical and electrical form, employ standard or nonstandard (when approved by the command or agency concerned) parts, and be completely representative of the final equipment.

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3.1.2 Materials. All materials shall be in accordance with MIL-E-16400, except as hereinafter specified.

3.1.2.1 The use of components or materials in the construction of the complementary filter of the combiner that have nonlinear characteristics (iron core coils, for example) is prohibited.

3.1.2.2 The basic structural material for chassis panel and enclosures shall be aluminum or aluminum alloy.

3.1.3 Selection of parts. Selection of all parts shall be as specified in MIL-E-16400.

3.1.3.1 Requests for approval for the use of nonstandard parts shall be submitted in strict accordance with the requirements of MIL-E-16400.

3.2 Composition. The combiner shall consist of the following:

(a) Complementary filter.
(b) Active receiver multicoupler.
(c) Power supply.
(d) Overload protection circuit and alarm.

3.2.1 Complementary filter. A complementary filter type of signal separation network shall be employed that shall separate incoming signals above and below the prescribed crossover frequency of 1.85 MHz. Signals below 1.85 MHz shall be passed through the low-pass section of the complementary filter and delivered to the active receiver multicoupler. Signals above 1.85 MHz shall be passed through the high-pass section of the complementary filter and delivered to existing 2 MHz to 30 MHz distribution systems.

3.2.2 Active receiver multicoupler. The active receiver multicoupler shall provide a prescribed amount of low-noise amplification and shall develop 8 isolated broadband outputs, each of which may be connected to a separate radio receiver operating in the 10 kHz to 1.85 MHz frequency range. Appropriate overload circuitry shall be provided to protect the active multicoupler from signals of distructive amplitudes. The overload circuitry shall be integral with the active multicoupler.

3.2.3 Power supply. An appropriate power supply shall be provided to develop the necessary operating voltages for the active multicoupler. All of its components shall be standard military parts. There shall be no non-standard parts as described in 3.1.3 incorporated in this section of the equipment.

3.2.4 Overload protection and alarm. An appropriate overload protection device with alarm shall be provided to meet the requirements of 3.7.6.2, 3.7.6.3, and 3.8.2.7.

3.3 Frequency range. The frequency range shall be inclusive from 10 kHz to 30 MHz. Signals below 1.85 MHz shall be delivered to the active receiver multicoupler. Signals above 1.85 MHz shall be delivered to the output for existing 2 MHz to 30 MHz distribution systems.

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3.4 Primary power source. The active receiver multicoupler shall operate from a 115 volt ± 10 percent, 60 Hz ± 5 percent, single-phase power source.

3.5 Orientation. The equipment shall be capable of storage in any orientation and shall be capable of operation in any angle up to 90 degrees from the normal.

3.6 Reliability.

3.6.1 Quantitative reliability requirement. The mean time between failure (MTBF), θ_o as defined by MIL-STD-781, of this equipment shall be 30,000 hours.

3.6.2 Reliability program plan. The supplier shall submit a reliability program plan in accordance with, and in the paragraph sequence of, MIL-STD-785 (see 6.3). Particular attention shall be paid to the sections of MIL-STD-785 that require submission of detailed tasks, procedures to implement and control these tasks, and time schedules. The proposed reliability program plan shall be submitted as a separate and complete entity within the total system or equipment project delineation. The reliability program plan as approved and incorporated into the contract becomes the basis for contractual compliance. The plan, including reliability test plans, must describe an integrated effort within the total project. It shall provide information to show how the supplier will meet specified quantitative reliability requirements during development and manufacture. The final reliability program plan shall be submitted for Government approval.

3.6.3 Reliability prediction. The contractor shall perform a reliability prediction in accordance with the paragraph of MIL-HDBK-217 entitled "How to Make a Reliability Prediction Based on Stress Factors and Part Populations". The prediction shall assume a 50°C ambient temperature and a shipboard installation. The contractor shall perform a thermal analysis and survey in accordance with MIL-T-23103 for purposes of determining thermal stress levels. For electrical stress levels, the manufacturer shall perform a theoretical electrical analysis to determine pertinent voltage and current levels in all circuits. This analysis shall be submitted to the government for approval.

3.6.3.1 Source of failure rates. The primary source of failure rate data shall be MIL-HDBK-217. For items not included in MIL-HDBK-217, other sources of data may be used with the approval of the command or agency concerned. When such alternate sources of data are proposed, enough information shall be submitted to enable the command or agency concerned to evaluate the validity and applicability of the results derived therefrom (see 6.3).

3.6.4 Reliability reports. Reports providing a complete accounting of progress on each element defined by the reliability program plan, results achieved, and status of actions to resolve problems and correct weak links shall be prepared for the command or agency concerned (see 6.3).

3.7 General requirements.

3.7.1 Weight and mounting dimensions. The weight of the combining network shall not exceed 50 pounds. The CU-2007( )/SRR shall be capable of being installed in a standard 19 inch rack (see MIL-STD-189), panel size GC.

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3.7.1.1 The overall dimensions shall not exceed the following:

Height	Width	Length
13 in.	19 in.	20 in.

3.7.2 Human engineering aspects of detail design. Equipment design shall conform to the criteria of MIL-STD-1472 and the requirements stated herein.

3.7.2.1 Front panel layout. A patch-panel arrangement to facilitate patching between any receiver and any coupler shall be provided on the front panel. The arrangement shall include 8 receiver terminals and 8 coupler terminals. Arrangement of the patching terminals shall be as shown in figure 1. Also the following controls and indicators shall be provided on the front panel:

(a) An ac power on-off switch that interrupts both sides of the ac line.
(b) An ac power indicator lamp.
(c) An overload indicator lamp.
(d) An overload reset control.

3.7.3 Removable assemblies. The complementary filter, active multicoupler and power supply shall be separate assemblies and individually removable from the front panel.

3.7.4 Cables and connectors.

3.7.4.1 All radio frequency terminals in the rear of the equipment shall be female coaxial, type C.

3.7.4.2 The 115 volt ac power terminal shall mate with a type MS 3106A14S7S cable plug.

3.7.4.3 An appropriate jumper cable shall be provided to make the connection between the low-pass output terminal of the complementary filter and the input terminal of the active multicoupler.

3.7.4.4 All terminals except those for front panel patching shall be accessible from the rear of the equipment.

3.7.4.5 The patch panel shall consist of:

(a) Twenty patch cords type TP PL-W4 1, or equal (see NAVSHIPS 0967-303-8610, Additional equipment supplied).
(b) Eight coupler terminations using receptacles that are compatible with patch cord type TP PL-4W 1, or equal. (See figure 2.)
(c) Eight receiver terminations routed from the rear receiver input terminals using receptacles identical to the coupler terminations. (See figure 2.)
(d) Four spare radio frequency (RF) terminations routed from the rear spare RF input terminals using receptacles identical to the coupler terminations. (See figure 2.)

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3.7.5 Fuses. Each side of the ac line shall be fused. Fuses shall be of the glass melamine type and shall be mounted in a fuse holder.

3.7.5.1 Blown fuse indicators, which conform to the requirements of MIL-F-19207, shall be provided for each active fuse.

3.7.6 Power-handling capability and linearity requirements.

3.7.6.1 Termination conditions. The high-pass output of the complementary filter shall be terminated in a 50 ohm dummy load. The low-pass output of the complementary filter shall be terminated in the active multicoupler.

3.7.6.2 Between 10 kHz and 2 MHz the power-handling capability shall be 10 watts root mean square (rms), 10 watts, peak envelope power (PEP), available power from a 50 ohm source.

3.7.6.3 Between 2 MHz and 30 MHz the power-handling capability shall be 500 watts rms, 5 kilowatts (kW) PEP.

3.7.6.4 Continuous operation at the power levels specified in 3.7.6.2 and 3.7.6.3 shall not degrade the performance below the requirements of section 3.

3.7.6.5 All intermodulation products that are between 2 MHz and 30 MHz, created by two 50 volt rms signals applied to the input terminal of the complementary filter at any two frequencies between 2 MHz and 30 MHz and detectable at the high-pass output terminal of the complementary filter, shall be 110 decibels (dB) or more below 50 volts. (See figure 4.)

3.7.6.6 All intermodulation products that are between 10 kHz and 1.85 MHz, created by two 50 volt rms signals applied to the input terminal of the complementary filter between 2 MHz and 30 MHz and detectable at any output terminal of the active multicoupler, shall be 110 dB or more below 50 volts. (See figure 5.)

3.7.6.7 All intermodulation products created by two 0.25 volt rms signals applied to the input terminal of the complementary filter between 10 kHz and 1.85 MHz shall be measured at 70 dB or more below 0.25 volt at any output of the active multicoupler. (See figure 5.) For purposes of reliability calculations required in 3.6, the power levels described in 3.7.6.2 shall be considered to occur 5 percent of the time. The power handling capability described in 3.7.6.3 shall be considered to occur 30 percent of the time. Intermodulation product levels shall not be degraded by impedance terminations on the remaining 7 unused outputs. This shall hold true for all of the 7 remaining short circuited, open circuited, or terminated outputs in any other combination of impedance levels.

3.7.7 Temperature. The CU-2007( )/SRR shall be capable of operation within the limits of class 4 of MIL-E-16400.

3.7.8 Shock, vibration, and inclination. The equipment shall meet the requirements of MIL-E-16400 and grade A, hull-mounted, class I, lightweight, type A shock requirements of MIL-S-901.

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3.7.9 Noise. The noise requirements of MIL-E-16400 are not applicable to this equipment.

3.7.10 Electromagnetic compatibility. This equipment shall conform to the electromagnetic compatibility requirements of MIL-STD-461 for class 1A equipment.

3.7.11 Phase relationship.

3.7.11.1 The phase relationship of the output signal at any of the 8 outputs of the active multicoupler to the input signal of the complementary filter for any frequency between 10 kHz and 65 kHz shall be constant within one-half microsecond (μs) as the temperature is varied over any 35°C segment of the operating temperature range.

3.7.11.2 In the frequency range between 10 kHz and 65 kHz the phase relationship of an input signal and the signal at any of the 8 outputs shall remain constant within one-half μs for any combination of termination applied to the remaining 7 outputs.

3.7.11.3 In the frequency range between 10 kHz and 65 kHz the phase relationship between an input signal and any of the 8 outputs shall remain constant within one-half μs for any termination impedance applied to the high-pass output of the complementary filter.

3.7.12 Enclosure and cooling. The enclosure shall be dripproof as specified in MIL-STD-108. The equipment enclosure as well as the equipment in general shall be assigned to make optimum use of the forced air cooling available in the CY-4516(A)/S equipment cabinet (see NAVSHIPS 0967-320-0010). The supplier shall demonstrate in the design review that this has been done. He shall also discuss in the equipment technical manual how the equipment is to be installed in the CY-4516(A)/S cabinet in order to achieve maximum cooling.

3.8 Detailed requirements.

3.8.1 Complementary filter.

3.8.1.1 Frequency response characteristics. The frequency response characteristics of the low-pass section, as measured in a 50 ohm system between the input terminal of the complementary filter and any output terminal of the active receiver multicoupler, shall provide the following:

(a) 2 dB ± 1 dB of gain between 10 kHz and 1.7 MHz.
(b) Variable from 2 dB ± 1 dB of gain at 1.7 MHz to not more than 5 dB of loss at 1.85 MHz.
(c) 50 dB or more of loss between 2 MHz and 30 MHz.

The above frequency response characteristics shall be met for any condition of termination applied to the high-pass output of the complementary filter and to the 7 outputs of the active multicoupler not under test.

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3.8.1.1.1 The frequency response characteristics of the high-pass section, as measured in a 50 ohm system between the input terminal of the complementary filter and the 2 MHz to 30 MHz output terminal of the complementary filter, shall provide the following:

(a) More than 20 dB of loss between 10 kHz and 1.7 MHz.
(b) Variable from 20 dB or more of loss at 1.7 MHz to not more than 6 dB of loss at 1.85 MHz.
(c) 0.5 dB or less of loss between 2 MHz and 32 MHz.

The above response characteristics shall be met for any condition of termination applied to the output of the active multicoupler.

3.8.1.2 Input impedance characteristics. The input impedance characteristics, as measured at the input of the complementary filter, with the low-pass output of the complementary filter terminated in the active multicoupler and the high-pass output of the complementary filter terminated in a 50 ohm dummy load, shall be as follows:

(a) Between 10 kHz and 1.85 MHz, the input voltage standing wave ratio (VSWR) relative to 50 ohms shall be less than 2.0:1.
(b) Between 2 MHz and 30 MHz, the input VSWR relative to 50 ohms shall be less than 1.3:1.
(c) Between 1.85 MHz and 2.0 MHz, the input VSWR relative to 50 ohms shall be less than 3.0:1.

3.8.2 Active multicoupler.

3.8.2.1 Number of outputs. The active multicoupler shall have 8 outputs.

3.8.2.2 Frequency range. The multicoupler shall operate over the frequency range of 10 kHz to 2 MHz.

3.8.2.3 Gain. The active multicoupler shall have a nominal gain of 2 dB ± 1 dB.

3.8.2.4 Noise figure. The noise figure of the multicoupler shall not exceed 9 dB.

3.8.2.5 Impedance (input and output). The input impedance shall be compatible with the requirements of 3.8.1.2. The output impedance of the active multicoupler shall be 50 ohms with a VSWR not to exceed 2.0:1.

3.8.2.6 Isolation.

3.8.2.6.1 The isolation between any 2 output terminals of the active multicoupler over the 10 kHz to 1.85 MHz frequency range shall be 30 dB or more.

3.8.2.6.2 The isolation between any output terminal and the input terminal of the active multicoupler over the 10 kHz to 1.85 MHz frequency range shall be 40 dB or more. This shall apply for any impedance termination of the remaining 7 outputs.

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3.8.2.7 Overload. An overload device shall be provided to protect the active multicoupler from the signal levels of destructive magnitude that would be encountered because of the power-handling requirements of 3.7.6.2. Operation of the overload device shall be initiated by detecting and responding to signal amplitudes in excess of a preset threshold level. Response time of the overload protection device shall be a maximum of .01 seconds. As specified, when the equipment is turned off, the overload circuit shall still the active receiver multicoupler.

3.8.2.7.1 The threshold level for overload protection shall be 0.7 volt ± 0.15 volt rms at the input terminal of the active multicoupler. Sustained operation at less than the threshold level shall not be destructive to the active multicoupler.

3.8.2.7.2 The signal level during overload, delivered to the 8 isolated broadband outputs (terminated in 50 ohms) throughout the power-handling conditions specified in 3.7.6.2, shall be less than 1 volt rms.

3.8.2.7.3 A signal lamp type of indicator and an audible alarm shall be provided to indicate an overload. The alarm shall be at a frequency of 2.9 ± 5 kHz and at a minimum sound level of 80 dB above 2 x 10^{-4} dynes/centimeter at 2 feet.

3.8.2.7.4 Reset of the overload device following an overload shall be accomplished by means of a manually operated push-button control located on the front panel.

3.8.2.7.5 There shall be no sustained noise or interference signals generated in the equipment during overload throughout the power handling required by 3.7.6.2. For purposes of reliability calculations required in 3.6, the overload protection and alarm shall be considered to operate once every 100 hours with the alarm turned on for 5 minutes each time.

3.9 Maintainability.

3.9.1 Quantitative maintainability requirements. The combiner shall have an equipment repair time (ERT) of no greater than 30 minutes when repair is accomplished by replacement of modules and parts. The ERT for the piece part repair of each reparable module shall be no greater than 60 minutes. ERT includes localization, isolation, disassembly, interchange, reassembly, alignment, and checkout for all modules and parts.

3.9.2 Qualitative maintainability requirements. The equipment shall be constructed to provide ease of maintenance, accessibility, and replacement of all modules and parts.

3.9.3 Plug-in construction. Optimum use shall be made of plug-in construction. The active receiver multicoupler, power supply, and overload protection as specified in 3.2 shall be plug-in modular construction.

3.9.4 Interchangeability. Unless otherwise approved by the command or agency concerned, mechanical and electrical interchangeability shall exist

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between similar assemblies, subassemblies, and replaceable parts supplied by a contractor. In the design of equipment, provisions shall be made for a design tolerance sufficient to accommodate, without departure from the specified performance, such articles as microcircuits, resistors, and other parts having their limiting dimensions and characteristics specified in the specifications for the particular part involved. The contractor shall prepare a list of all interchangeable items.

3.9.5 Maintainability program. The contractor shall prepare and submit for approval to the command or agency concerned a maintainability program as specified in MIL-STD-470 (see 6.3). The maintainability program shall describe the tasks to be performed and the procedures for conducting and controlling the maintainability program. The program shall address each paragraph of MIL-STD-470 in sequence and shall define and identify the interface between the maintainability and integrated logistic support requirements.

3.9.6 Maintainability prediction. A maintainability prediction shall be performed using Procedure II, Part A, Corrective Maintenance, of MIL-HDBK-472. The contractor shall document and justify all assumptions and the applicability of all data used in the prediction. The prediction shall include all mechanical, electromechanical, and electronic parts. When repair is accomplished by module replacement, the prediction shall be prepared in two parts. Part I of the prediction shall be performed for module and part replacement. Part II shall be accomplished for the piece part repair of all reparable modules. The prediction shall be compared with the required ERT, and whenever the predicted values do not meet the requirements, the contractor shall accomplish the changes in design that are necessary to improve the predicted values sufficiently to meet the requirements. The prediction shall be submitted for approval to the command or agency concerned (see 6.3).

3.9.7 Maintainability analysis. A maintainability analysis shall be performed concurrently with the design effect for the incorporation of the quantitative and qualitative maintainability requirements into the equipment design. The maintainability analysis shall evaluate equipment design and design changes and the possible catastrophic and critical modes of equipment failure. The analysis shall be submitted for approval to the command or agency concerned (see 6.3).

3.9.8 Maintainability demonstration tests. The maintainability requirements of the equipment shall be demonstrated as specified in 4.6.

3.9.9 Test equipment requirements.

3.9.9.1 Circuit test points. Circuit test point accesses that comply with the criteria outlined in MIL-STD-415 shall be provided. The following types of tests shall be considered for determining the test points: (a) signal quality, (b) circuit performance, (c) calibration, (d) system alignment, (e) installation and checkout, (f) planned maintenance systems tests, (g) line replaceable unit (LRU) fault location, and (h) shop replaceable part (SRP) fault isolation.

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3.9.9.2 Support test equipment. The selection of general purpose test equipment (GPTE) shall be determined by the following tests: (a) signal quality, (b) circuit performance, (c) alignment, (d) tuning, (e) peaking, (f) adjustments, (g) calibration, and (h) LRU fault location.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Inspection system requirement. The contractor shall provide and maintain a written inspection system acceptable to the Government for supplies and services covered by this specification. The inspection system shall be in accordance with MIL-I-45208.

4.1.2 Quality plan. A quality plan applicable to the equipments covered by this specification shall be prepared for Government approval (see 6.3).

4.1.3 Government verification. All quality assurance operations performed by the contractor will be subject to Government verification at any time. Verification will include (a) surveillance of the operations to determine that practices, methods, and procedures of the written inspection system are being properly applied; (b) Government product inspection to measure quality of product to be offered for acceptance; and (c) Government product inspection of delivered products to assure compliance with all inspection requirements of this specification. Failure of the contractor to correct promptly deficiencies discovered by him or of which he is notified shall be cause for suspension of acceptance until corrective action has been made or until conformance of product to prescribed criteria has been demonstrated.

4.1.4 Test procedures. The contractor shall prepare test procedures for Government approval covering first article and quality conformance inspection (see 6.3).

4.1.5 Quality assurance terms and definitions. Quality assurance terms used in this specification shall be as defined in MIL-STD-109.

4.2 Classification of inspection. The method of examination and testing shall be classified as follows:

(a) First article inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

(1) Production inspection (see 4.4.1).
(2) Production control inspection (see 4.4.2).
(3) Environmental inspection (see 4.4.3).
(4) Inspection for preparation for delivery (see 4.4.4).

(c) Reliability testing (see 4.5).
(d) Maintainability demonstration (see 4.6).

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4.3 First article inspection. Unless otherwise specified (see 6.2), one unit shall be required for first article inspection. First article inspection shall consist of all examination and testing necessary to determine compliance with the requirements of this specification. In addition to the listed tests of MIL-E-16400, first article inspection shall include the inspections specified in table I.

Table I. First article inspection.

Inspection	Reference
Production inspection	4.4.1
Production control inspection	4.4.2
Environmental inspection	4.4.3
Nonstandard parts approval	3.1.3.1
Thermal survey	3.6.3
Cables and connectors	3.7.4
Blown fuse indicator (compliance)	3.7.5.1

4.3.1 First article inspection report. A first article inspection report shall be submitted to the Government covering the results of the contractors first article inspection (see 6.3).

4.4 Quality conformance inspection.

4.4.1 Production inspection. Production inspection shall be made on all equipment offered for delivery. The inspection shall comprise such examination and testing as will prove workmanship and reveal omissions and errors of the production process. Tests shall include functional and performance tests at a limited number of points, tests that detect deviations from design, tests of adjustment, and tests that detect hidden defects of material. Production inspection shall consist of the tests and examinations that are approved as specified in 4.1.4 and shall include the requirements of table II.

Table II. Production inspection.

Inspection	Reference
Visual	
Materials	3.1.2
Enclosure material	3.1.2.2
General requirements	
Frequency range	3.3
Weight and mounting dimensions	3.7.1 and 3.7.1.1
Ac power on-off switch	3.7.2.1

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Table II. Production inspection. (Continued)

Inspection	Reference
Ac power indicator lamp	3.7.2.1
Fuses	3.7.5
Blown fuse indicators (functioning)	3.7.5.1
Power-handling capability	3.7.6.2 and 3.7.6.3
Phase relationship	3.7.11
Detailed requirements	
Complementary filter	
Frequency response characteristics	3.8.1.1 and 3.8.1.1.1
Input impedance characteristics and VSWR	3.8.1.2
Active multicoupler	
Frequency range	3.8.2.2
Gain	3.8.2.3
Noise figure	3.8.2.4
Overload alarm and indicator lamp	3.8.2.7.3
Overload reset control	3.8.2.7.4

4.4.1.1 Intermodulation. The contractor shall demonstrate conformance to 3.7.6.5, 3.7.6.6, and 3.7.6.7 by performing adequate testing at the frequencies given in table III. A method of performing the intermodulation measurement is presented in 4.4.1.1.1.

Table III. Production inspection intermodulation.

Input Test Signal	f_1 (MHz)	f_2 (MHz)	Intermodulation Test Freq. for Testing to 3.7.6.5 (MHz)	3.7.6.6 (MHz)	3.7.6.7 (MHz)
50 volts each	2.000	3.000	4.0, 7.0, 8.0	1.00	
	2.000	3.9000	5.8	0.10	
	2.000	3.990	9.98	0.01	
	0.01	0.035			0.015, 0.025, 0.045 0.055, 0.060, 0.080
0.25 volts each	0.10	0.35			0.15, 0.25, 0.45 0.55, 0.60, 0.80
	0.80	1.10			0.30, 0.50, 1.40

4.4.1.1.1 Intermodulation measurements. As shown in figure 3, two signals, f_1, and f_2, between 2 MHz and 30 MHz are filtered, combined, and delivered to a linear 50 ohm termination. The levels of f_1 and f_2 are adjusted individually for 50 volts rms. A path to a spectrum analyzer or receiver is provided by way

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of a decoupling pad and a bandpass filter tuned to the intermod frequency of interest. The decoupling pad and the bandpass filter, f_3, isolate the receiver or spectrum analyzer from the fundamental energy at f_1 and f_2. The instrumentation is checked first to determine the residual intermodulation level at the intermodulation product of interest. The complementary filter and active multicoupler are then inserted into the measurement system as shown in figure 4. The 50 volt test levels are reestablished at the input of the complementary filter. Measurements are then made of the intermodulation level at the frequencies of interest. Figure 4 is a diagram of intermodulation testing as specified in 3.7.6.5. Figure 5 is a diagram of the test instrumentation for 3.7.6.6 and is valid for the requirements of 3.7.6.7 except that power-type signal sources are not necessary. Any intermodulation distortion in the measurement system and dummy load shall be below the specified requirement for the complementary filter.

4.4.2 Production control inspection. Production control inspection, including sampling, shall conform to table IV and to the inspection procedures of MIL-STD-105 using the special inspection levels. The inspection level shall be S-3 for normal, tightened, and reduced inspection. Production control inspection shall be performed on equipments that have passed production inspection. Production control inspection shall be approved in accordance with 4.1.4. The equipment shall satisfactorily meet the requirements of production control inspection prior to release for shipment.

Table IV. Production control inspection.

Inspection	Reference	AQL%
Primary power source	3.4	6.5
Overload	3.7.2, 3.8.2.7, 3.8.2.7.1, 3.8.2.7.2, 3.8.2.7.3, 3.8.2.7.4, and 3.8.2.7.5	6.5
Linearity requirements	3.7.6.4	6.5
Enclosure	3.7.12	6.5
Impedance and VSWR	3.8.2.5	6.5
Isolation	3.8.2.6	6.5
Intermodulation	4.4.1.1	6.5
Power	MIL-E-16400	6.5

4.4.2.1 Rejected lots. If an inspection lot is rejected, the supplier may withdraw the lot from further inspection. The supplier may also rework a rejected lot to correct the defective units and reinspect the lot using tightened

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inspection. Rejected lots shall be kept separate from new lots and shall not lose their identity.

4.4.3 Environmental inspection. Environmental inspection shall be as listed in table V and shall be performed on units that have been subjected to, and have passed, production inspection. Environmental tests shall be approved in accordance with 4.1.4.

Table V. Environmental inspection.

Inspection	Reference
Shock, vibration, and inclination	3.5 and 3.7.8
Humidity	MIL-E-16400
Temperature	3.7.7
Electromagnetic compatibility	3.7.10

4.4.3.1 Sampling for environmental inspection. One equipment shall be selected from each successive 100 equipments produced or fraction thereof. The first sample equipment shall be selected from the first month's production.

4.4.3.2 Nonconforming environmental sample units. If a sample unit fails the inspection specified in 4.4.3, the contractor shall immediately investigate the cause of failure and shall report to the quality assurance representative the results thereof and details of the corrective action taken to correct units of product that were manufactured under the same conditions, with the same materials, processes, and so forth. If the quality assurance representative does not consider that the corrective action will enable the product to meet specified requirements, or if the contractor cannot determine the cause of failure, the matter shall be referred to the contracting officer (see 6.5).

4.4.3.3 Reinspection of conforming environmental sample units. Unless otherwise specified in the contract, sample units that have been subjected to, and have passed, environmental tests may be accepted on the contract provided they are resubjected to, and pass, production inspection after repair of all damage.

4.4.4 Inspection for preparation for delivery. Inspections shall be conducted to ensure conformance to the requirements of section 5.

4.5 Reliability test procedures. Using eight CU-2007( )/SRR combiners, the contractor shall prepare reliability test procedures that are in accordance with test level C, test plan XXV of MIL-STD-781. No voltage cycling or on-off cycling is required (see 6.3).

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4.5.1 Test setup. During reliability testing the equipment shall be operated as follows:

(a) For 3 hours each day, a 500 watt rms signal shall be applied in the 2 MHz to 30 MHz frequency range. The first frequency shall be 2 MHz and shall increase by 1 MHz each day up to 30 MHz. This cycle shall be repeated throughout the reliability test.
(b) For 21 hours each day, the equipment shall be operated in the 300 kHz to 1 MHz frequency range with a 0.25 volt rms signal. The power supply shall be operated for 24 hours a day.

4.5.2 Daily measurements. Once each day the following tests shall be performed:

(a) 500 kHz to 1 MHz range - amplifier gain, power supply voltage, and visual inspection.

4.5.2.1 The following measurements shall be made at 25 days and 50 days after the test has begun:

(a) Intermodulation (see 3.7.6).
(b) Isolation (see 3.8.2.6).
(c) Noise figure (see 3.8.2.4).

4.5.2.2 Frequency response measurements, in accordance with 3.8.1.1, shall be made at 12, 25, 37, and 50 days after the test has begun.

4.5.3 Disposition of tested equipment. Upon satisfactory completion of the reliability test, the equipment shall be subjected to the production inspection of 4.4.1 verifying compliance with 3.7.6 through 3.8.2.7.5. After these tests, the equipment shall be refurbished as necessary and submitted as part of the production quantity provided the equipment meets all contractual requirements.

4.5.3.1 If a reject decision is reached, the action required by MIL-STD-781 shall be taken.

4.5.4 Reliability sampling test. Six out of every 100 units or fraction thereof produced shall be randomly selected by the cognizant Government inspector for reliability testing in accordance with 4.5 through 4.5.3.1.

4.6 Maintainability demonstration test.

4.6.1 Maintainability demonstration. Conformance to the maintainability requirements shall be demonstrated in accordance with Method 3 of MIL-STD-471. The maintainability demonstration shall be completed successfully prior to the delivery of any production equipment and prior to the start of reliability testing.

4.6.2 Maintainability demonstration test plan. The contractor shall prepare a detailed maintainability test plan in accordance with MIL-STD-471. The maintainability test at the organizational level shall be performed by a

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technician with experience equivalent to that of a third-class naval electronics technician (ET-3). The plan shall include provisions to demonstrate the ERT's for the equipment. The equipment maintainability requirements (30 minute ERT) shall demonstrate by replacement of modules and parts at the organizational level. Technical documentation to be used shall be limited to the equipment technical manual. If the technical manual is not available, a preliminary draft shall be substituted upon approval of the command or agency concerned. The maintainability demonstration test plan shall be submitted to the command or agency concerned prior to the start of the test (see 6.3).

4.6.3 Maintainability demonstration report. A maintainability demonstration report shall be prepared in accordance with MIL-STD-471 (see 6.3).

4.7 Support maintenance test procedure verification (prime equipment).

4.7.1 The procedures and techniques developed to perform all types of maintenance tests for each level (organizational, intermediate, depot, special repair facilities, and calibration facilities) shall be verified prior to submitting: (a) final data of MIL-STD-1345(EC), (b) Navy training instructions, and (c) final Navy technical manuals.

4.7.2 The support maintenance test procedure verification shall be performed by a technician selected by the Government from the contractor or Government personnel.

4.7.3 The test procedures used in the support maintenance test procedures verification may be preliminary procedures, pamphlets, or manuals (not necessary in the required technical manual format) or preliminary manuals in the standard format.

4.7.4 Test equipment used to verify the support maintenance test procedures shall be selected from MIL-STD-1364, Preferred General Purpose Electronic Test Equipment. Test equipment not specified in MIL-STD-1364 may be used in the support maintenance test procedure verification only if the contractor submits a written request to the command or agency concerned and approval is granted (see 6.3).

4.7.5 The support maintenance test procedure verification shall be performed on the first article and one equipment selected from the production run.

4.7.6 Support maintenance test procedure verification.

(a) A technician shall be given a complete combiner network that is disassembled to its lowest line replaceable part.
(b) The technician shall install (reassemble) the modules or boards in the prime equipment and align the prime equipment as specified in the manual.
(c) The technician shall operate the equipment as specified in the manual (all modes) and verify that the equipment meets all specification requirements such as: input and output impedances, noise figure, isolation, overload, power handling, and linearity requirements.

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5. PREPARATION FOR DELIVERY

(The preparation for delivery requirements specified herein apply only for direct Government procurements. Preparation for delivery requirements of referenced documents listed in section 2 do not apply unless specifically stated in the contract or order. Preparation for delivery requirements for products procured by contractors shall be specified in the individual order.)

5.1 Preservation, packaging, packing, and marking. The preparation for delivery requirements to be called for in the procurement documents shall be in accordance with the applicable levels specified in MIL-E-17555 and MIL-STD-129 (see 6.2).

6. NOTES

6.1 Intended use. This equipment is intended for use with surface ship antennas and receivers.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Number of first article samples if other than specified in 4.3.
(c) Preservation, packaging, packing, and marking required (see 5.1).

6.3 Contract data requirements. Data generated by this document are not deliverable unless specified on the Contract Data Requirements List (DD Form 1423) or the contract schedule. The data required by this specification include, but are not restricted to, the following:

(a) Reliability program plan (see 3.6.2).
(b) Reliability prediction (see 3.6.3).
(c) Source of failure rates (see 3.6.3.1).
(d) Reliability reports (see 3.6.4).
(e) Maintainability program (see 3.9.5).
(f) Maintainability prediction (see 3.9.6).
(g) Maintainability analyses (see 3.9.7).
(h) Quality plan (see 4.1.2).
(i) Test procedures (see 4.1.4).
(j) First article inspection report (see 4.3.1).
(k) Reliability test procedures (see 4.5).
(l) Maintainability demonstration test plan (see 4.6.2).
(m) Maintainability demonstration report (see 4.6.3).
(n) Test equipment selection request (see 4.7.4).

6.4 First article. Invitations for bids should provide that the Government reserves the right to waive the requirement for first article samples for those bidders offering a product that has been previously procured or tested by the Government and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending procurement.

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6.5 Environmental inspection. Approval to ship may be withheld at the discretion of the Government, pending the decision from the contracting officer on the adequacy of corrective action.

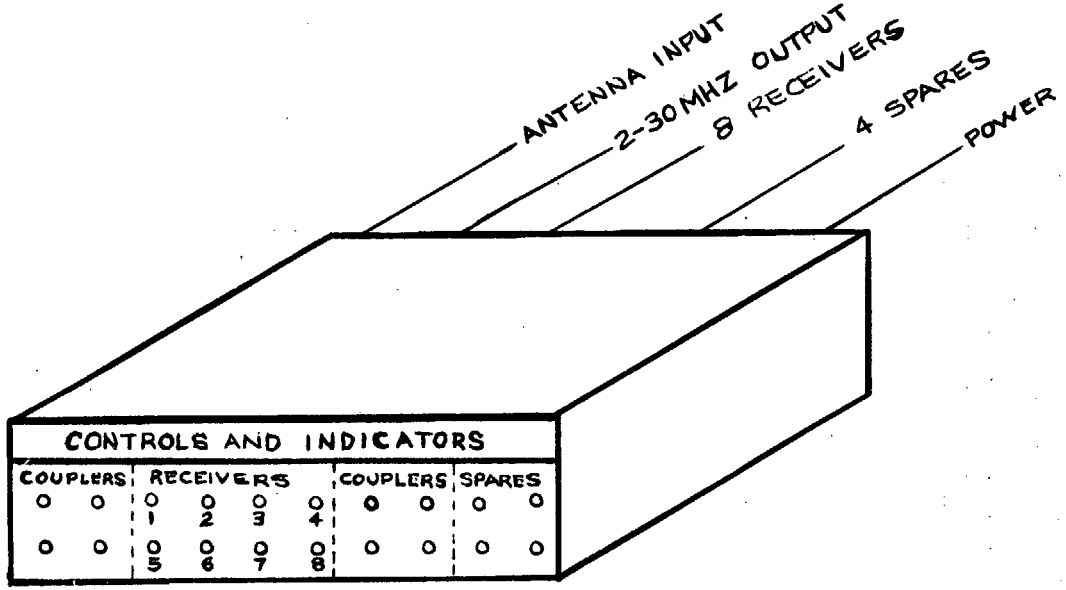
6.5.1 Initial shipments. Initial shipment of end items shall be contingent upon successful completion of the environmental inspection.

6.6 Changes from previous issue. The outside margins of this document have been marked "#" to indicate where changes (deletions, additions, and so forth) from the previous issue have been made. This has been done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content as written irrespective of the marginal notations and relationship to the last previous issue.

Preparing activity:
Navy - EC

(Proj. No. 5985-N376)

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NOT TO SCALE

Figure 1. CU-2007( )/SRR front panel layout. (Terminals shall be numbered as shown.)

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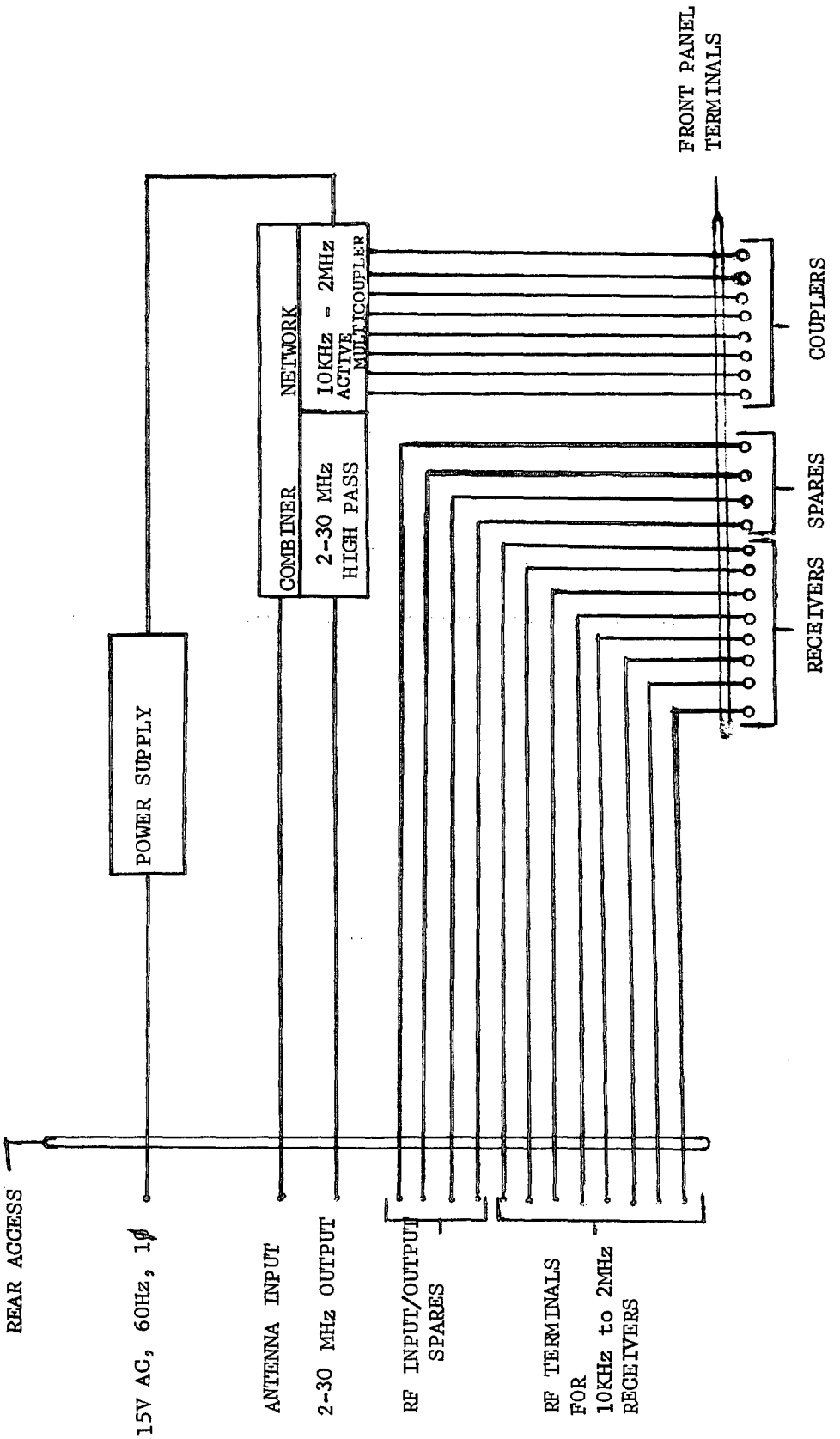


Figure 2. CU-2007( )/SRR wiring diagram for rear access and front panel patching.

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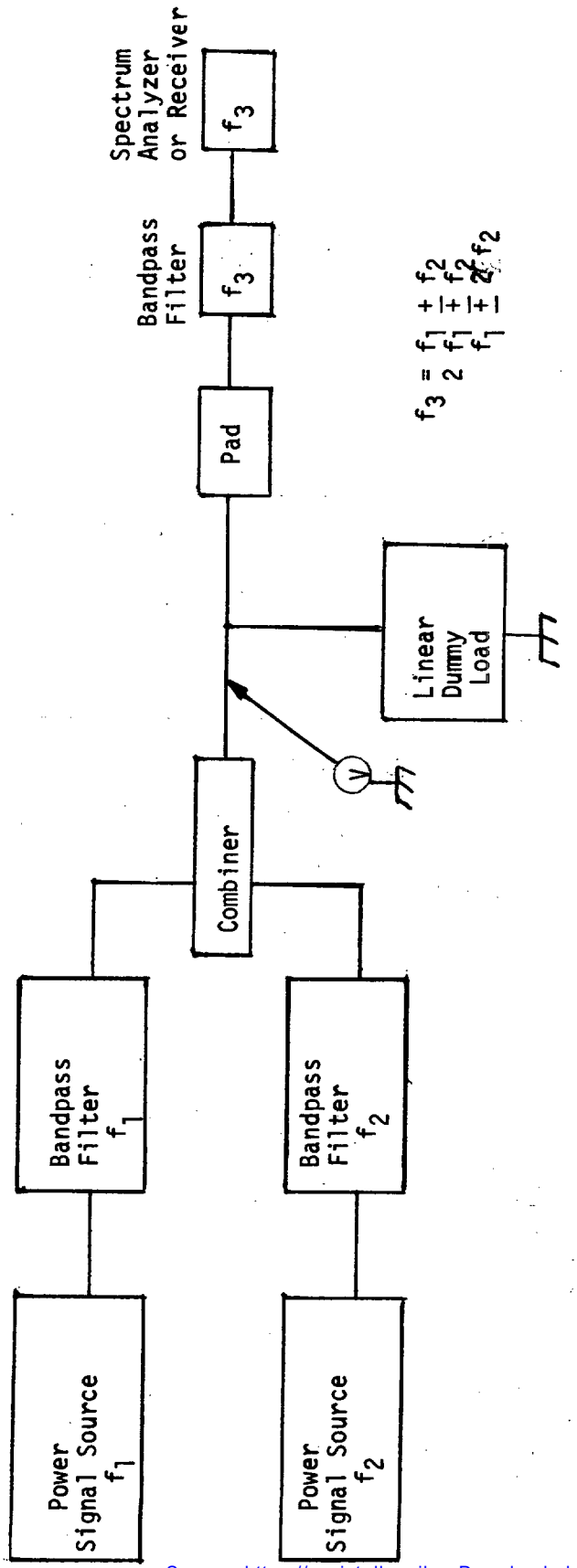


Figure 3. Instrumentation for system intermodulation measurement.

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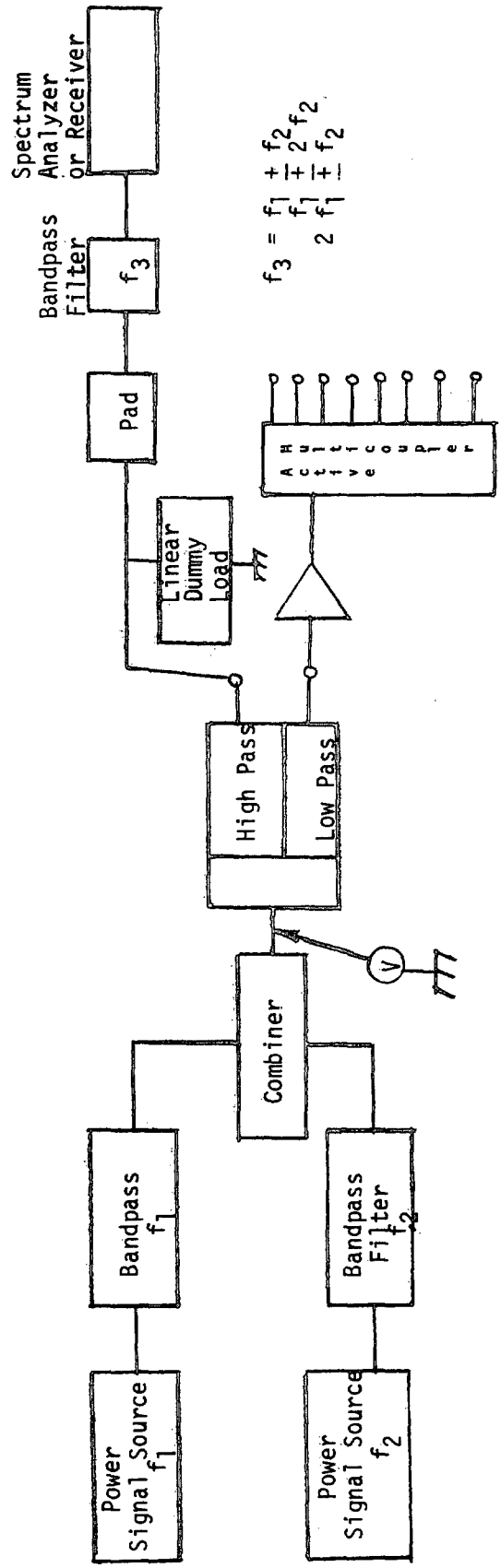


Figure 4. Intermodulation testing as specified in 3.7.6.5.

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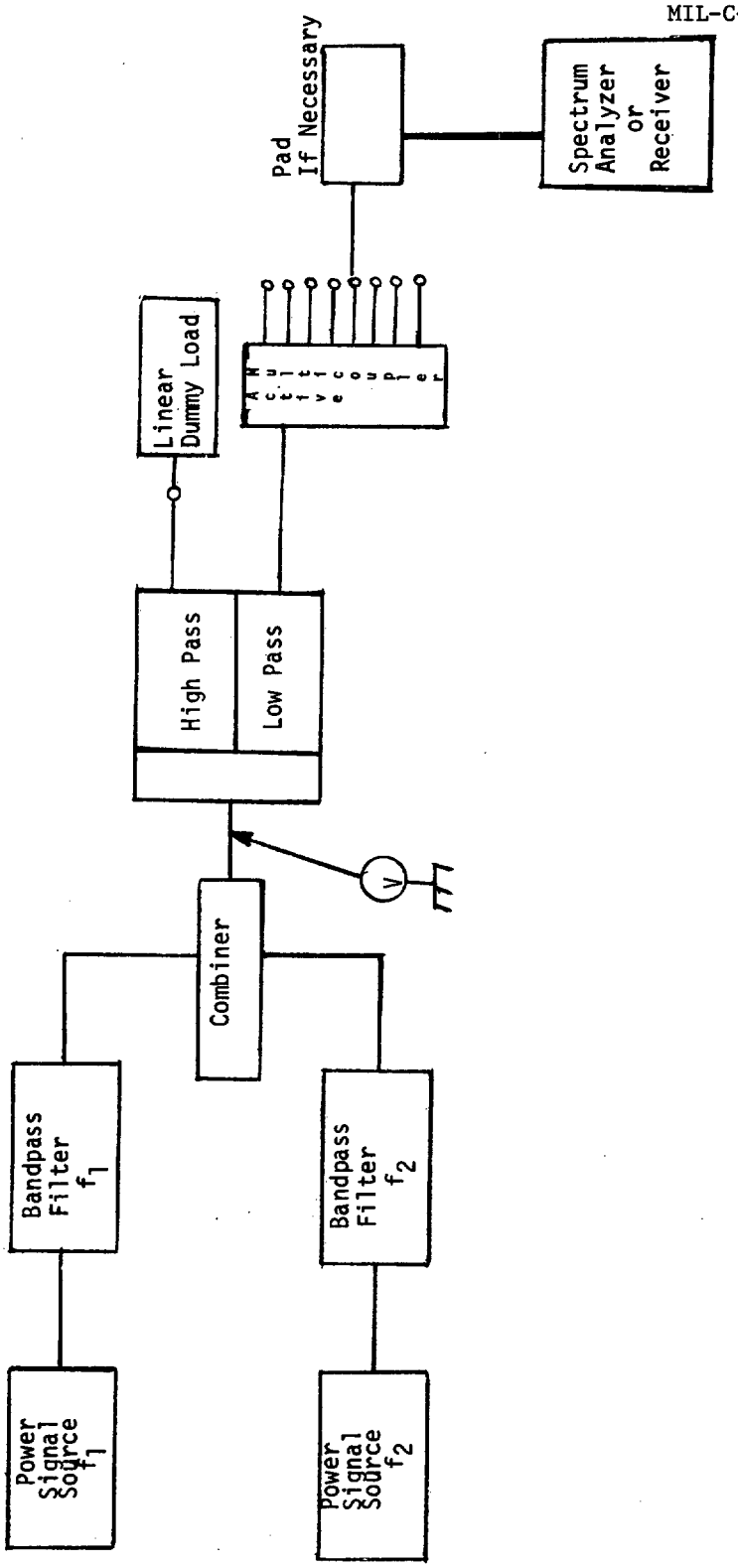


Figure 5. Test instrumentation for 3.7.6.6 (also for 3.7.6.7 except that power signal sources are not necessary).

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426 REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

S/N-0102-014-1801 C-25254

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